

Artificial Intelligence Automated Cybersecurity Checking Tool.

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for this threat modelling project I am setting a target AI system, mapping possible threats and proposing mitigations for the possible threats.

1. My target AI system

Human Resource Chatbot

2. Mapping Possible Threats

MITRE ATLAS™

Matrix

Tactics

Techniques

Mitigations

Case Studies

Works in research or controlled settings, but is not necessarily demonstrated in realistic operations.

Contribute

Reconnaissance

Resource Development

Initial Access

AI Model Access

Execution

Persistence

Privilege Escalation

Defense Evasion

Credential Access

Discovery

Lateral Movement

8 techniques

13 techniques

7 techniques

4 techniques

6 techniques

9 techniques

4 techniques

15 techniques

6 techniques

9 techniques

2 techniques

Active Scanning	Acquire Infrastructure	AI Supply Chain Compromise	AI Model Inference API Access	AI Agent Clickbait	AI Agent Context Poisoning	AI Agent Tool Invocation	AI Supply Chain Reputation Inflation	AI Agent Tool Credential Harvesting	Cloud Service Discovery	Phishing
Gather RAG-Indexed Targets	Acquire Public AI Artifacts	Drive-by Compromise	AI-Enabled Product or Service	AI Agent Tool Invocation	AI Agent Tool Data Poisoning	Escape to Host	AI Supply Chain Rug Pull	Credentials from AI Agent Configuration	Discover AI Agent Configuration	Use Alternate Authentication Material
Gather Victim Identity Information	Develop Capabilities	Evade AI Model	Full AI Model Access	Command and Scripting Interpreter	AI Agent Tool Poisoning	LLM Jailbreak	Corrupt AI Model	Exploitation for Credential Access	Discover AI Artifacts	
Search Application Repositories	Establish Accounts	Exploit Public-Facing Application	Physical Environment Access	Deploy AI Agent	LLM Prompt Self-Replication	Valid Accounts	Delay Execution of LLM Instructions	OS Credential Dumping	Discover AI Model Family	
Search Open AI Vulnerability Analysis	LLM Prompt Crafting	Phishing		LLM Prompt Injection	Manipulate AI Model		Evade AI Model	RAG Credential Harvesting	Discover AI Model Ontology	
Search Open Technical Databases	Obtain Capabilities	Prompt Infiltration via Public-Facing Application		User Execution	Modify AI Agent Configuration		Exploitation for Defense Evasion	Unsecured Credentials	Discover AI Model Outputs	
Search Open Websites/Domains	Poison Training Data	Valid Accounts			Poison Training Data		False RAG Entry Injection		Discover LLM Hallucinations	
Search Victim-Owned Websites	Publish Hallucinated Entities				Prompt Infiltration via Public-Facing Application		Impersonation		Discover LLM System Information	
	Publish Poisoned AI Agent Tool				RAG Poisoning		LLM Jailbreak		Process Discovery	
	Publish Poisoned Datasets						LLM Prompt Obfuscation			
	Publish						LLM Trusted Output Components Manipulation			
							Manipulate Host LLM			

Mapping Possible Threat Table

System Layer	Threat	Mitigations
Input Layer	Prompt Injection (malicious query override instructions)	Input filtering, regex and guardrails
LLM Processing Layer	Data Leakage (LLM Reveals Sensitive Data)	Safety classifications, restricted system prompts, limited context window
HR Document Database	Unauthorised DB access (retrieving payroll/SSN data)	Role based access control, encryption at rest
Output Layer (Chatbot Response)	Hallucinations (false or misleading answers)	Output filtering, human to human loop for sensitive queries
System Infrastructure	API Misuse (DoS, Insecure logging, weak authentication)	API rate limits, secure logging

3. Proposed Mitigations in the scenario of an “AI Human Resource Chatbot”